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1

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product performance from 3 years of transaction data to guide strategic decisions.

2

Dataset at a Glance

3,900 Purchases

18 Columns

37 Missing Ratings

get Gender, Location, Subscription Category, Amount, Season, Size, etc., Frequency, Rating, Shipping

3

Data Preparation in Python

1 Load & Explore Imported with pandas, used df.info() and df.describe()

2 Clean & Impute Filled missing Review Ratings with category median

3 Engineer Features Created age_group and purchase_frequency_days columns

4 Load to PostgreSQL Connected Python to PostgreSQL for SQL analysis

4

Revenue & Spending Insights

Shipping Type vs. Spend

Shipping	Avg. Spend
Standard	\$78.46
Express	\$85.48

85% discount users still spent above average high-value segment.

5

Top-Rated Products

Most Discounted Products

Item	Discount Rate
Hat	55.00%
Sneakers	45.00%
Coat	45.00%
Sweater	45.00%
Pants	45.00%

6

Customer Segmentation

Subscribers vs. Non-Subscribers

Status	Customers	Avg. Spend	Revenue
Subscriber	1,253	\$79.49	\$60,440
Non-Sub	2,647	\$79.87	\$170,430

Repeat buyers (1+ purchases): 95% subscribed vs. 45% not subscribed.

Top Products by Category

Clothing

Accessories

Like what you created?

+ Create something else

↶ Back to prompt

Help refine our product

How satisfied are you with the output?

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